

Investigation of significant capacity recovery effects due to long rest periods during high current cyclic aging tests in automotive lithium ion cells and their influence on lifetime



Bernd Epding^{a,*}, Björn Rumberg^a, Hannes Jahnke^a, Ina Stradtman^a, Arno Kwade^b

^a Volkswagen AG, Group Research, Letterbox 011/11720, 38436 Wolfsburg, Germany

^b Technische Universität Braunschweig, Institute for Particle Technology, Volkmaroder Straße 5, 38104 Braunschweig, Germany

ARTICLE INFO

Keywords:

Lithium ion cell testing
Long rest periods
Lithium plating
Capacity recovery
Microscopic modeling

ABSTRACT

In this work for automotive lithium ion cells, the influence of prolonged rest periods of up to 5 days on the available capacity and the long term aging trend during cyclic aging tests is presented. Automotive prismatic cells with NMC/graphite chemistry and over 25 Ah capacity are investigated by means of various cycle tests. The test matrix includes a variation of possible influencing factors like current rates during cycling as well as temperature and state of charge during rest periods, rest period length and number of cycles between rest periods.

A significant capacity recovery during rest periods is measured after cycling at moderate temperatures which has only been reported for low temperature lithium plating experiments before. Rest periods were conducted at 100% state of charge, thus the anode overhang effect can be excluded as the cause for the observed capacity increase. Voltage relaxation tests indicate that lithium plating occurs due to high current rates. Furthermore, a microscopic model is proposed to explain the long time constant of the limited exponential capacity increase during rest periods. Under these plating conditions a significant increase in cell cycle lifetime is observed if regular rest periods or standard characterizations are applied.

1. Introduction

With the development of more and more hybrid electric and electric vehicles (HEV, EV) which utilize lithium ion batteries, a thorough understanding of their aging behavior is essential. So far the lifetime of lithium ion batteries was only a minor concern due to the short innovation cycles in consumer electronics. However, for the use in EVs lifetime is a key factor and research has already determined the most important causes for capacity loss and impedance rise [1]. According to this, simulations and models to predict battery lifetime often only focus on these loss factors [2]. Due to the fast pace of innovation in battery cell technology these models become most vital to quickly assess if a new generation of batteries will endure years of usage in EVs in all possible scenarios. The models are usually trained based on a limited number of accelerated aging tests (calendar and cyclic) which must not take longer than a few months to keep up with automotive development timelines. So far aging tests are performed over a period of one year or longer. This is too long and expensive and the aim is to reduce the necessary test time to 6 months or less.

An important but often neglected fact is that privately owned cars

spend only a small fraction of their lifetime driving. Long, continuous rest periods of hours (at night, while at work) or even days (over the weekend) are part of most realistic usage scenarios. However, the effect of rest periods on the lifetime of lithium ion cells and respective cell tests has hardly been investigated. It is necessary to understand the physical effects of rest periods to be able to translate accelerated aging tests to realistic usage scenarios.

So far Reichert et al. investigated the effect of short rest periods of up to 2 h on the aging behavior of various types of 18,650 type lithium ion cells [3]. They found no significant difference in the aging behavior for nickel manganese cobalt oxide/graphite (NMC/C) or lithium iron phosphate/graphite (LFP/C) based cells if cycled without any rest periods or with 2 h rest periods in between cycles. However, they found an exponentially limited capacity increase of up to 0.6% during rest periods of up to 5 days after 5 cycles at 20 °C.

Furthermore, Eddahech et al. studied the influence of 2 day rest periods on the aging behavior of 12 Ah NMC polymer cells [4]. Their cells were tested using an adapted IEC 62660-1 driving cycle to compare continuous usage with a 5 day usage/2 day rest scenario. The cells were cycled at 45 °C and 55 °C and there was an observable increase in

* Corresponding author.

E-mail address: bernd.epding@volkswagen.de (B. Epding).

<https://doi.org/10.1016/j.est.2019.02.015>

Received 6 December 2018; Received in revised form 11 February 2019; Accepted 11 February 2019

Available online 22 February 2019

2352-152X/ © 2019 Elsevier Ltd. All rights reserved.

Table 1

Cell test parameters of cells referenced in this work. Reference number and test parameters are given as well as production batch.

Cell No.	Charge/ Discharge rate	Cycles between rest periods	Rest period length	Temperature during cycling	Temperature during rest periods	SoC during rest periods	Batch
1	2C/1C	–	–	10	–	–	2
2	2C/1C	50	2	10	10	100	2
3	2C/1C	100	2	10	10	100	2
4	1C/2C	–	–	10	–	–	1
5	1C/2C	100	Char. *	10	23	–	1
6	1C/2C	50	2	23	10	100	1
7	2C/1C	50	2	23	10	100	1
8	2C/1C	50	2	23	23	100	1
9	2C/1C	50	0.5/1/2/5 repeat	23	23	100	1

*Char. stands for a characterization procedure with electrochemical impedance spectroscopy where the cell is brought to a reference temperature and is cycled four times with 1C. During the test the cell has multiple prolonged rest periods to equilibrate temperatures.

usable cell capacity after the 2 day rest period. A dependency on the state of charge (SoC) during the rest period was also found. However, due to the nature of the complex test scenario a quantitative analysis of the capacity recovery effect is difficult. More recently the attention turned towards the effect that the geometrical anode overhang has on the measured capacity of lithium ion cells. The anode is normally geometrically oversized compared to the cathode to prevent lithium plating on the edge of the anode. Furthermore, wound cells often contain a large anode overhang facing outwards or inwards at the end of the jelly roll. It was at first described by Gyenes et al. to explain coulombic efficiency measurements greater than 100% [5]. The effect was further explored by Lewerenz et al. and Wilhelm et al. who studied how the anode overhang affects calendar aging tests [6–9]. They discovered that the excess anode material in the geometrical anode overhang influences capacity measurements. When a cell is stored for prolonged time at a high SoC, lithium ions in the charged part of the anode which is directly opposite to the cathode can move into the anode overhang. This is driven by the potential difference between the lithiated particles in the active anode area and delithiated particles in the overhang. Since the lateral distances to the graphite particles in the overhang can vary between mm and even cm, the charging and discharging of the overhang area can take hours or even many days. This can affect the measured capacity significantly since the size of the passive anode can be up to 9% [9].

Moreover, Lewerenz et al. have recently shown that the homogeneity of lithium distribution (HLD) in the anode can decrease during shallow cyclic aging [8,10]. This decreases the usable cell capacity due to the cell reaching its voltage limits earlier. Following the cyclic aging the usable cell capacity increased for their cells during calendar storage at the same SoC as during cycling over a period of 100–200 days. This increase in usable capacity was linked to an increase in HLD.

Finally, Petzl et al. and Waldmann et al. have observed an increase in usable capacity when they applied “regeneration cycles” or rest periods at higher temperatures after forcing lithium plating at low temperatures [11,12]. Both observed a limited exponential increase in usable capacity over the course of a few days. However, to the best of our knowledge there is no analytically proven microscopic model of how this slow reversible process works.

Due to the nature of accelerated aging tests for aging models, defined high current rates are applied to the cells to achieve a large number of cycles and thereby high charge throughput quickly. Under many of these conditions lithium plating is very likely to occur.

In this work it was investigated how prolonged rest periods in long term high current cyclic aging tests affect the behavior of the tested automotive cells. This observed effect has to be taken into account if quality aging models need to be parameterized by accelerated aging tests. A strong dependency of the impact of rest periods on the cell test conditions is observed. Regular rest periods or characterizations during accelerated cyclic aging tests lead to an increase in the measured cell capacity as well as to a longer lifetime of the cells. A capacity increase is

measured even if the cell is left at 100% SoC during the rest period. This excludes the anode overhang as the source of the capacity increase. The effect is further linked to high charge current rates where lithium plating occurs even at higher temperatures. Furthermore, the capacity rise depends exponentially on the rest period duration and is hardly influenced by temperature variations from 10 °C to 23 °C. An explanation for this behavior is proposed to be the reactivation of passivated metallic lithium plating. As lithium plating is a self-enforcing process, more frequent rest periods, in which lithium plating is removed, reduce the capacity loss during cycling. The gathered data indicates that during 2 day rest periods about 40% of lithium plating can be recovered.

2. Experimental

In order to gain insight into the effect of longer rest period durations and to be able to quantify the effect of rest periods depending on various utilization parameters, a series of cell tests was performed. Automotive prismatic cells with NMC/graphite chemistry and over 25 Ah capacity were used. These cells are delivered at 30% SoC. The cells are braced to constant distance between two aluminum plates with 0.4 Nm per screw as recommended by the manufacturer. This increases the cycle lifetime as shown by Wünsch et al. and drastically improves the reproducibility of the cycle tests [13]. Only cells from the same production batch are compared. Within one production batch these highly automatically produced cells are very reproducible (see supporting information). An overview of the different test parameters is given in Table 1. Note that these cycle tests do not include any standard characterization procedure unless explicitly stated. This is because most standard characterization procedures contain prolonged rest periods for temperature and charge equilibration, especially before measuring electrochemical impedance spectroscopy. As will be shown below, these characterizations can lead to a significantly different long term aging trend than cycling without characterizations.

All cell tests were carried out in Binder and Vötsch climate chambers (± 1 °C). Kratzer and Digatron test systems were used to cycle the cells and measure their capacity. During the prolonged rest periods the cells remained in their climate chamber which was set to the temperature as given in Table 1.

Full cycles are composed of a constant current – constant voltage (CCCV) charge step to a maximum voltage of 4.15 V which is terminated at a cut off current of $I < \frac{C_{nom}}{20}$ and a CC discharge step down to 3 V. Between charge and discharge steps there was a rest period of 10 min.

The usable cell capacity (UCC) is used to be able to compare different cells under identical test conditions:

$$UCC = \frac{C_{cyc,i}}{C_{cyc,1}} * 100\% \quad (1)$$

Where $C_{cyc,1}$ and $C_{cyc,i}$ are the cell capacities as measured in the first respectively i 's cycle under test conditions. This means the capacity of

each cycle is normalized to the first cycle under identical test conditions. Thus, UCC must not be confused with state of health (SoH), which would be determined by a standard characterization procedure.

The capacity change during rest periods (ΔC_{rest}) is calculated as the difference between the capacity of the second cycle (to avoid measuring self-discharge during the rest period) after the rest period and the capacity of the last cycle before the rest period relative to the capacity at the start of the experiment:

$$\Delta C_{rest} = \frac{C_{cyc,k+2} - C_{cyc,k}}{C_{cyc,1}} * 100\% = UCC_{k+2} - UCC_k \quad (2)$$

In this case k denotes the last cycle before the rest period.

In order to determine whether lithium plating occurs at a given charge rate and temperature voltage relaxation C-rate tests as described by Schindler et al. were conducted [14]. To this end, the cell was brought to a desired temperature and discharged with 1 C to 3 V. Afterwards, the cell was charged with constant currents with varying C-rates (C/10 to 3 C) until the cut-off voltage of 4.15 V was reached. Then, the cell voltage was measured every second for up to two hours in open circuit condition. According to literature a shoulder appears in the relaxation voltage curve if lithium plating occurred in the charge step due to the coexistence of lithium metal (0 V vs. Li/Li⁺) and graphite (0.08 V vs Li/Li⁺) on the anode side [14]. Finally, the cell was discharged with 1 C and the test was repeated with the next C-rate.

3. Results and discussion

The test matrix as given in Table 1 was designed to understand how prolonged rest periods of up to 5 days influence cell behavior during cyclic aging tests. At first, it is shown that rest periods and characterization periods can significantly influence cell cycle life. Furthermore, the interdependencies of various test parameters with respect to rest periods are examined. Here the focus is on the occurrence of lithium plating during cyclic aging tests. A microscopic model is proposed to explain the long time constant of the capacity recovery during rest periods. Finally, it is shown that a significant amount of previously lost cell capacity can be recovered during 2 day rest periods under various test conditions.

3.1. Influence of rest periods on cell cycle lifetime

In order to analyze the influence of prolonged rest periods on the long term aging trend of lithium ion cells, tests were performed to compare cycling with and without defined rest periods.

Fig. 1a) depicts how 2 day rest periods every 50th or 100th cycle

change the aging behavior of a lithium ion cell when compared to a cell test without any rest periods at all nor any characterizations. All three cells were cycled at 2 C charge, 1 C discharge current at 10 °C. Fig. 1a) shows that a cell ages most quickly when there are no rest periods allowed during cycling. The cell reaches 80% UCC after about 330 full cycles. A cell which was paused fully charged for 2 days every 100th cycle already gains another 40 full cycles and reaches 80% UCC after 370 full cycles. Finally, the cell with rest periods at full charge every 50th cycle lasts for a whole 560 cycles, almost doubling the required number of cycles to reach 80% UCC.

Fig. 1b) compares cell tests conducted at 10 °C with 1 C charge and 2 C discharge currents. Cell 4 was cycled continuously without any rest periods. Cell 5 was characterized at 23 °C every 100 cycles which includes longer rest periods for temperature equilibration and four full cycles with 1 C charge and discharge rate. Up until 300 cycles it can be argued that cell 5 ages similarly compared to cell 4 as the capacity approaches the capacity of cell 4 during cycling between rest periods. However, after about 300 cycles the characterizations seem to cause cell 5 to age much slower than cell 4. It takes cell 5 another 270 additional cycles to reach 85% UCC compared to cell 4.

The presented data leaves us to believe that rest periods or characterization periods which include prolonged rest times, respectively, can have a significant impact on long term cyclic aging tests of lithium ion cells. Especially, if such tests are used to parameterize empirical cell models, predictions of realistic usage scenarios with longer rest periods (car is idle most of the day) or continuous cycling can be erroneous. To be able to make proper predictions the underlying physical effects of aging and recovery during rest periods in lithium ion cells need to be understood.

To this end, this report attempts to work out the circumstances under which rest periods strongly influence cell tests and which mechanism causes the recovery phenomenon during rest times.

3.2. SoC during rest periods

The influence of the storage SoC during rest periods has already been discussed in literature [5,7–9]. The so called anode overhang (excess anode material on a geometrically larger anode sheet) can store lithium ions the same way that the active anode material does which is directly opposite to the cathode. A small overhang is influencing capacity measurements at a timescale of hours and a large overhang can influence capacity measurements over many days [5,7–9].

However, we point out that the theory of the anode overhang only explains a capacity rise if the SoC during a rest period is lower than the average SoC during the cycles before the rest period. In this case lithium

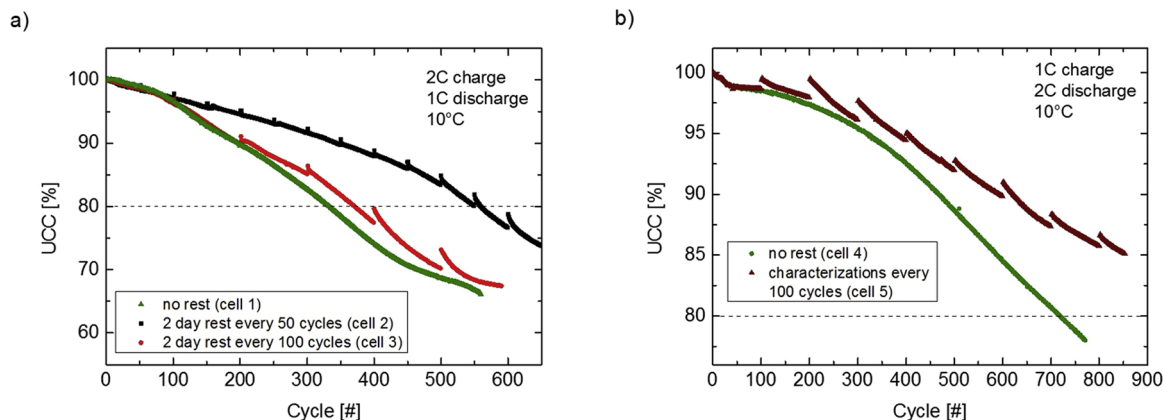


Fig. 1. (a) Aging of cells with varying number of cycles between rest periods. All cells were cycled at 10 °C with 2 C charge and 1 C discharge current. Rest periods were conducted at 100% SoC and 10 °C. (b) Comparison of cells with and without standard characterizations every 100 cycles. The cells were charged with 1 C and discharged with 2 C at 10 °C. Characterizations include change of temperature to 23 °C, electrochemical impedance spectroscopy, prolonged rest periods for equilibration and four full cycles at 1 C.

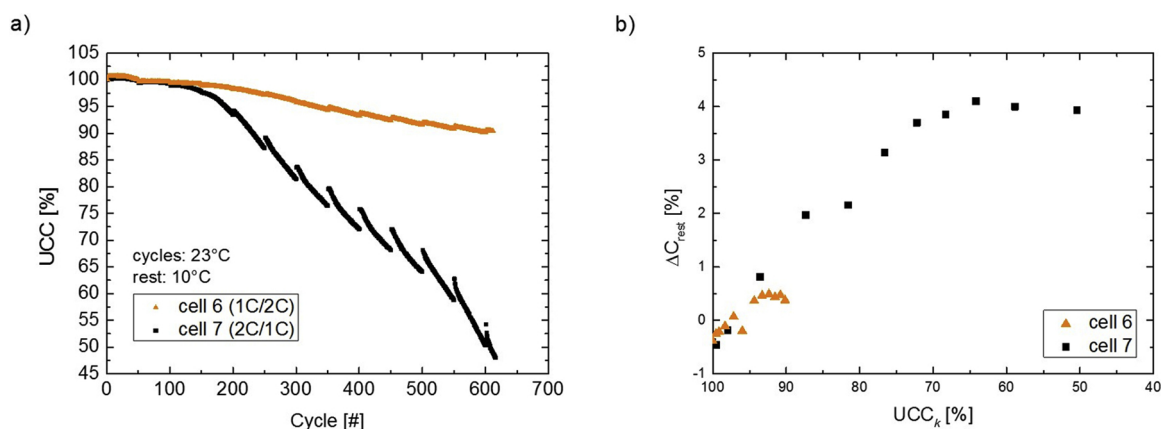


Fig. 2. Comparison of cells cycled at 1 C charge, 2 C discharge (cell 6) respectively 2 C charge, 1 C discharge (cell 7) current rate. Cycles were conducted at 23 °C while 2 day rest periods every 50th cycle were at 10 °C. In a) aging of both cells over cycle number, in b) capacity rise during rest periods over usable capacity before the rest period is shown.

stored in the overhang would diffuse back into the active part of the anode. Here the cells were delivered with a SoC of 30% and cycled with 100% depth of discharge. The anode is about 12% geometrically larger than the cathode whereas about 8% is the small overhang of the anode along the whole length of the electrode and 4% is the large overhang at the ends of the jelly roll (compare other prismatic cells in [9]). Tests are presented which were paused at 100% SoC. At this SoC lithium ions are expected to diffuse into the anode overhang and decrease the measured capacity. Still under certain test conditions the cells show a capacity recovery during rest periods at 100% SoC. This is the opposite effect as predicted by the theory of the anode overhang. Thus, there must be at least one additional process as is discussed in the following sections.

3.3. Influence of charge/discharge rate

In order to investigate the impact of the applied charging or discharging rate on the behavior during rest periods, various cells were cycled at different C-rates with two day rest periods every 50 cycles. In Fig. 2 a cell test for a 1 C charge, 2 C discharge rate (cell 6) is shown compared to a test with 2 C charge, 1 C discharge current rate (cell 7). Both cells were cycled at 23 °C and rest periods were conducted at 100% SoC at 10 °C. Fig. 2a) demonstrates that cell 7 behaves very similarly to cell 6 up to 150 cycles. Afterwards, cell 7 ages more quickly than cell 6. In Fig. 2b) the capacity increase during rest periods is shown over remaining usable capacity before each rest period. Both cells show a capacity loss during the first rest periods. However, after losing some overall capacity, the cells start to regain capacity during the following rest periods. This behavior is much more pronounced for cell 7 which is charged with a 2 C current rate compared to cell 6 which is charged with a 1 C current rate. It seems that an increase from 1 C to 2 C charging rate introduces or enforces an additional aging mechanism that leads to faster degradation and is partly reversible during rest periods. Lithium plating (metallic deposition of lithium on the graphite anode) is the most promising explanation. Even though the cycles were conducted at moderate temperatures of 23 °C, a 2 C charging rate can already lead to lithium plating especially after a certain aging.

In order to check if indeed lithium plating is occurring, a voltage relaxation test as described by Schindler et al. was conducted [14]. The test procedure is described in section 2. For this test a fresh cell of batch 2 was used (not shown in Table 1).

The data presented in Fig. 3 indicates that there is no lithium plating occurring at 23 °C up to a charging rate of 0.75 C in a fresh cell. However, at 1 C charging rate the voltage curve starts to show a “shoulder”, i.e. a turning point at about 200 s after the rest period starts. This shoulder is even more pronounced at 2 C and 3 C charging rate. According to literature the shoulder appears due to the coexistence of

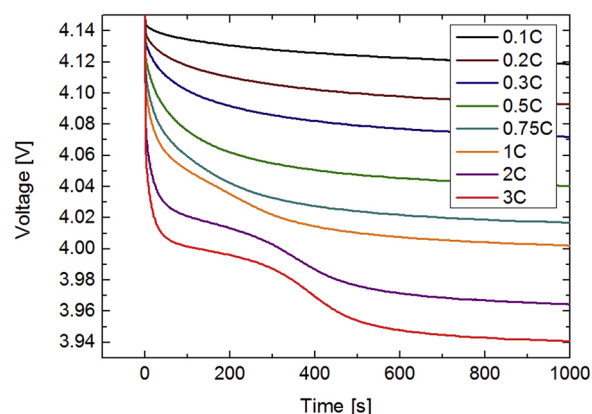


Fig. 3. Cell voltage during rest periods after charging at 23 °C with different constant current rates (from C/10 to 3 C). Above 0.75 C the voltage curves show a shoulder which indicates the occurrence of lithium plating in the previous charging step.

lithium metal (0 V vs. Li/Li^+) and graphite (0.08 V vs. Li/Li^+) on the anode side [14]. The shoulder disappears after a certain time since lithium metal will readily intercalate into the graphite particles as long as there is an electrical contact.

These results can explain the vastly different behavior observed during the cyclic aging tests presented in Fig. 2. At 1 C charging rate only minimal lithium plating occurs and thus, the cell loses its capacity very slowly. During the rest periods there is hardly any recovery as there is nothing to recover. The capacity loss in this case is most likely due to irreversible SEI (solid electrolyte interface) growth and reversible loss of lithium into the anode overhang [1,15]. At 2 C charging rate, however, there should be plenty of lithium plating during each cycle. This will lead to an increased capacity loss per cycle as lithium plating will damage the SEI and react with the electrolyte where it breaks through the SEI or lose electrical contact to the graphite in the following discharge [16,17]. Yet, we believe that during rest periods some of the previously immobilized metallic lithium plating, which lost contact to the graphite, can reestablish contact and again participate in cell reactions in the following cycles as described in section 3.6.

3.4. Dependency on rest period duration and cell age

In Fig. 4 the capacity change during each rest period for cell 9 is plotted over the rest periods duration. The cell was cycled with 2 C charge and 1 C discharge currents at 23 °C. Rest periods with varying duration were conducted at 100% SoC at 23 °C every 50th cycle. Each

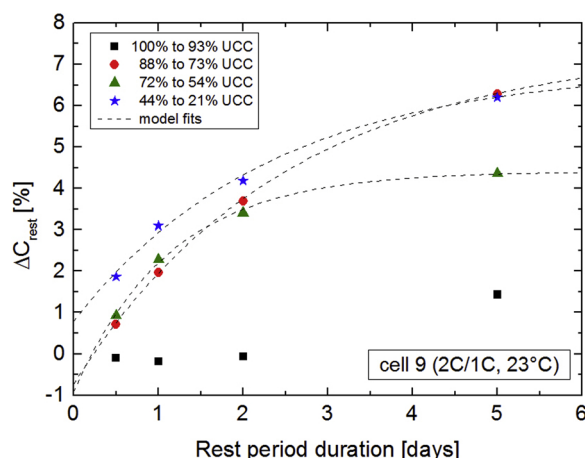


Fig. 4. Capacity increase during each rest period vs. rest period duration for cell 9. Cycles were performed with 2 C charge and 1 C discharge current rate. The test was performed at 23 °C. Rest periods were conducted at 100% SoC and their length was varied from 0.5, 1, 2 to 5 days. Each symbol (color) represents one sequence of 200 cycles out of the overall 1000 cycles with a rest period every 50 cycles. With every sequence the cell ages so that the UCC-range is given for every sequence in the legend.

symbol (color) in Fig. 4 represents one sequence of 200 cycles out of the overall 1000 cycles. The rest period length every 50th cycle was varied from 0.5, 1, 2 to 5 days during each sequence. This means after 50 cycles 0.5 days rest, after 100 cycles 1 day rest, after 150 cycles 2 days rest, after 200 cycles 5 days rest. After 250 cycles 0.5 days rest and so on. With every sequence the cell ages so that the UCC-range is given for every sequence in the legend. During the first three rest periods there is no capacity increase observed. The cell even loses some capacity which is why ΔC_{rest} is negative. This is likely caused by irreversible SEI formation as well as reversible lithium loss into the anode overhang. Since the anode overhang can become electronically disconnected when the cell ages, this effect should decrease over the lifetime of the cell [18]. Only the first 5-days rest period after aging to about 93% UCC results in an increase of the cells capacity. Below 90% UCC all rest periods result in a significant increase of cell capacity. With longer rest periods the capacity increases more, following a limited exponential behavior. This trend is almost identical to the effect observed by Waldmann et al. who studied the effect of a rest period at 23 °C after forcing lithium plating at low temperature in 18,650 cells [12]. The dependency of the capacity increase on rest period length varies over the lifetime of the cell but the general trend stays very similar below 90% UCC.

The relatively long time it takes to recover the capacity is quite interesting and we believe it cannot be explained alone by already known ways of reintercalation of metallic deposited lithium as described in literature. Zinthe et al. argue that it should take only about two hours to reintercalate plated lithium into the graphite anode at room temperature [19]. In the voltage relaxation tests shown in Fig. 3 the voltage plateau ascribed to the occurrence of metallic lithium on the anode even vanished after a few minutes (see Fig. 3). Additionally, most of metallic plated lithium should be oxidized during the next discharge and should not at all depend on a previous rest time. Thus, we are looking for a process that reactivates immobile lithium, which is lost most likely due to plating and cannot simply be oxidized during discharge. In Section 3.6 a microscopic model is presented how this long time constant for the reactivation process can be explained.

3.5. Temperature during rest periods

Since a temperature activated process was suspected to be responsible for the capacity increase during rest periods, the temperature during the rest period was varied while keeping it constant for the cells

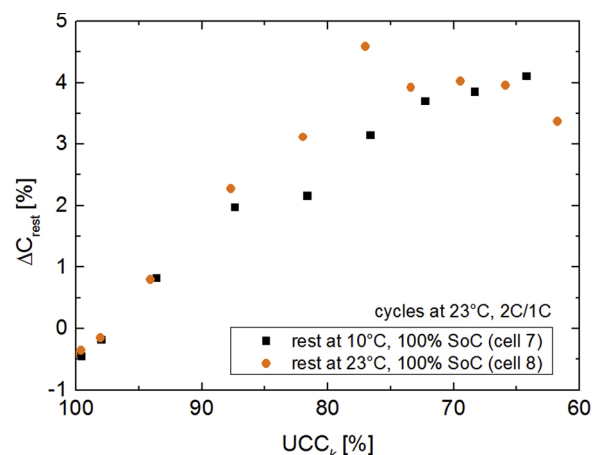


Fig. 5. Capacity increase after 2 day rest periods depending on temperature during the rest periods. The cycles between the rest periods were carried out under identical conditions. Note: The data point at 77% UCC_k for cell 8 is higher due to an unplanned longer rest period.

during cycling. The capacity change during rest periods for these cells, presented exemplary for cells 7 and 8 in Fig. 5 does not show a significant dependency on the temperature during the rest periods. On the contrary, both cells show a similar capacity recovery during 2 day rest periods. Fig. 4 shows that a 2 day rest period at 23 °C does not recover all the capacity. For a rest period at 10 °C a much slower capacity recovery should be expected. Thus, the amount of recovered capacity seems to be limited mainly by the rest time and is not accelerated or slowed down significantly by a 13 °C temperature change. This is interesting as any chemical reaction or diffusion process should be significantly sped up with an increase in temperature of 13 °C. Considering this result, the HLD as the main cause for the recovery of usable capacity during the 2 day rest periods shown here appears unlikely due to two reasons. First, the capacity increase correlated to an increase in HLD takes much longer than 2 days as shown by Lewerenz et al. [10]. Second, as an increase in HLD should depend on diffusion and intercalation reactions, an increase in rest period temperature should speed up the capacity recovery. Nonetheless, an increase in HLD is likely superimposed and responsible for a part of the observed capacity recovery.

3.6. Microscopic model for slow lithium plating reactivation

In Fig. 6 a possible microscopic picture is shown how a reactivation of previously immobilized metallic lithium on the graphite anode might occur. It is assumed that metallic lithium which is in good electrical contact with the graphite particles will quickly intercalate into them. This is because this situation is equivalent to discharging a graphite/lithium cell. Depending on temperature it takes only a few minutes until this process is finished and all metallic lithium is intercalated into the graphite [19]. However, after multiple cycles during which lithium plating occurred, some metallic lithium might be randomly passivated from all sides by a passivating layer while still having mechanical contact with the graphite particle [16]. This passivating layer should be very similar in form and composition to the SEI which forms on the graphite during formation [20]. Thus, it should be permeable to lithium ions while being a very poor electron conductor. In order for the metallic lithium to intercalate into the graphite and hence be accessible during the following cycles an electron from the metallic lithium has to move through the passivating layer to the graphite. The lithium ion left behind can then easily follow the electron like a lithium ion during normal charging. Hence, the poor electrical conductivity of the passivating layer which encapsulates the metallic lithium leads to the long time constant for this process [21]. This can also explain why an

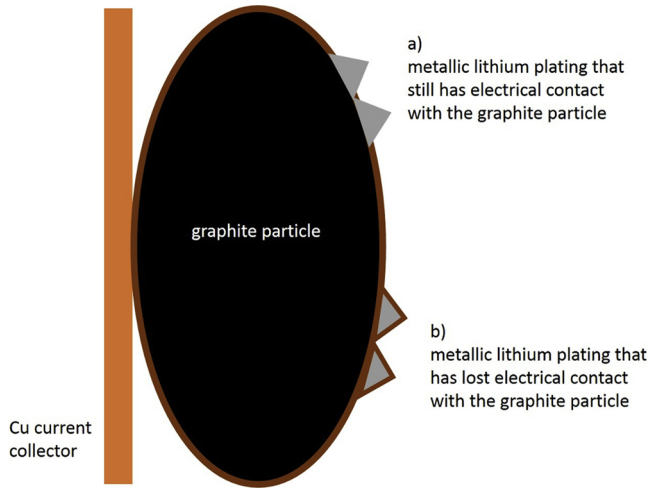


Fig. 6. Microscopic model to explain the very slow capacity increase during long rest periods after lithium plating occurred. In a) the metallic lithium still has electrical contact to the graphite particle. It can thus quickly intercalate into the graphite (shorted half-cell) or be oxidized during the next discharge. In b) the metallic lithium is still on the surface of the graphite but is passivated by a poorly electron conducting layer [16]. In order for a lithium ion to be able to intercalate into the graphite its electron has to move through the insulating layer first. The lithium ion can then follow as the passivating layer is permeable to lithium ions.

increase of the temperature during the rest period does not significantly speed up the capacity increase. The electron migration through the passivating layer should not be influenced significantly by a temperature increase of 13 °C. To formulate a mathematical model which can be fit to the experimental data it is assumed that the metallic lithium in Fig. 6b) is composed of N_{Li} lithium atoms which in turn have N_e valence electrons. The rate at which electrons move through the passivating layer into the graphite is proportional to the number of valence electrons left in the metallic lithium and a crossing frequency f :

$$-\frac{dN_e}{dt} = fN_e \quad (3)$$

This equation can be integrated to give:

$$N_e(t) = N_{e,0} \exp(-ft), \quad (4)$$

where $N_{e,0}$ is the initial number of valence electrons in the metallic lithium. Since the lithium ions will quickly follow their electrons once they crossed into the graphite, it can be assumed that the time development for the lithium ions will be the same as for the electrons:

$$N_{Li}(t) = N_{Li,0} \exp(-ft) \quad (5)$$

The measureable recovered cell capacity, which is the difference between the amount of lithium initially stored in metallic form and the number of lithium ions that moved into the graphite, will then follow the same time dependency:

$$C(t) = C_{max} (1 - \exp(-ft)) \quad (6)$$

Here C_{max} is the maximum capacity that could be recovered by an infinitely long rest period. To compensate for measurement inaccuracy and capacity loss during rest periods due to irreversible reactions of the metallic lithium with the electrolyte and loss of lithium into the anode overhang, an offset C_{off} is introduced:

$$C(t) = C_{off} + C_{max} (1 - \exp(-ft)) \quad (7)$$

Note that Eq. (7) can be readily converted to the equation which Waldmann et al. used to fit their data and thus offers a microscopic explanation for the very good fit results [12].

Eq. (7) was fitted to the data in Fig. 4 to obtain the values in Table 2. The results indicate that the microscopic description is valid for the

Table 2

Results from fitting Eq. (7) to the experimental data of the time evolution of the capacity during rest periods in Fig. 4.

	C_{off} [%]	C_{max} [%]	f [day ⁻¹]	R^2
88% to 73% UCC	-0.73 ± 0.08	8.13 ± 0.10	0.40 ± 0.02	0.999
72% to 54% UCC	-0.92 ± 0.40	5.33 ± 0.36	0.88 ± 0.12	0.992
44% to 21% UCC	0.79 ± 0.48	6.13 ± 0.50	0.43 ± 0.13	0.983

whole UCC range. However, the data is best fit when the cell is still relatively fresh (88% to 73% UCC). In this case the results are also a very close match to the data presented by Waldmann et al. [12], which they obtained from completely different cells. When the cell is aged further the microscopic situation might become more and more disturbed which limits the applicability of the presented model. Additionally, the data points for each fit were obtained at different UCC states. In order to further improve the experimental data, multiple cells would need to be aged to the same UCC and then paused for different rest times. This might mitigate the jumps in the parameter values.

The fact that C_{off} becomes positive for the data of the lowest UCC range might be explained by a very thick SEI that thoroughly passivates the graphite particles and is not easily pierced by lithium plating. This would limit the capacity loss during the rest period. Furthermore, the anode overhang might get electronically disconnected due to frequent volume changes during cycling which limits the reversible lithium loss into the anode overhang during the rest periods [18].

3.7. Number of cycles between rest periods

In order to understand the impact of the number of cycles between two rest periods cell tests were conducted where one cell was paused for 2 days after every 50th cycle while another cell was paused for 2 days every 100th cycle. Both cells were tested in the same climate chamber and the temperature was held at 10 °C throughout the whole test time (cycles and rest periods).

In Fig. 7a) the long term aging trend is shown for both cells. Cell 3 loses capacity faster than cell 2. The increase of usable capacity during every rest period is shown for both cells plotted over cycle number (b) and over remaining usable capacity before the respective rest period (c). Fig. 7b) suggests that more frequent rest periods strongly influence the behavior of lithium ion cells under plating conditions. Cell 2 which had rest periods every 50th cycle only starts to show a capacity recovery after about 300 cycles while cell 3 recovers capacity already after 200 cycles. Interestingly, the capacity increase after 2 day rest periods is comparable if plotted against the remaining usable capacity before the rest period (Fig. 7c)). Thus, it seems that C_{max} and f are only depending on the conditions at a certain UCC if the dominating aging mechanism under these test conditions is lithium plating.

It can be assumed that frequent rest periods allow a lithium ion cell to recover from a negative effect that lithium plating caused beforehand. An increasing sensitivity to lithium plating over lifetime has been discussed before by Ecker et al. and Schuster et al. [22,23]. An explanation for this behavior could be the clogging of lithium diffusion paths in the porous anode which in turn increases the chance of lithium plating in every cycle as suggested by theoretical investigations [24].

Since lithium plating is thought to be a self-enforcing process due to the clogging of pores in the porous anode or the formation of a thick covering layer on the anode surface [22,24–26], the frequent recovery during rest periods might suppress to some extent the strong aging effect that lithium plating can have. Also, a regular increase in HLD should have a positive effect on the aging behavior as inhomogeneities are increasing the risk of lithium plating. From the time dependence it is known that the recovery takes more than 2 days to settle. Thus, a longer regular rest time might negate the self-enforcing effect of plating even further.

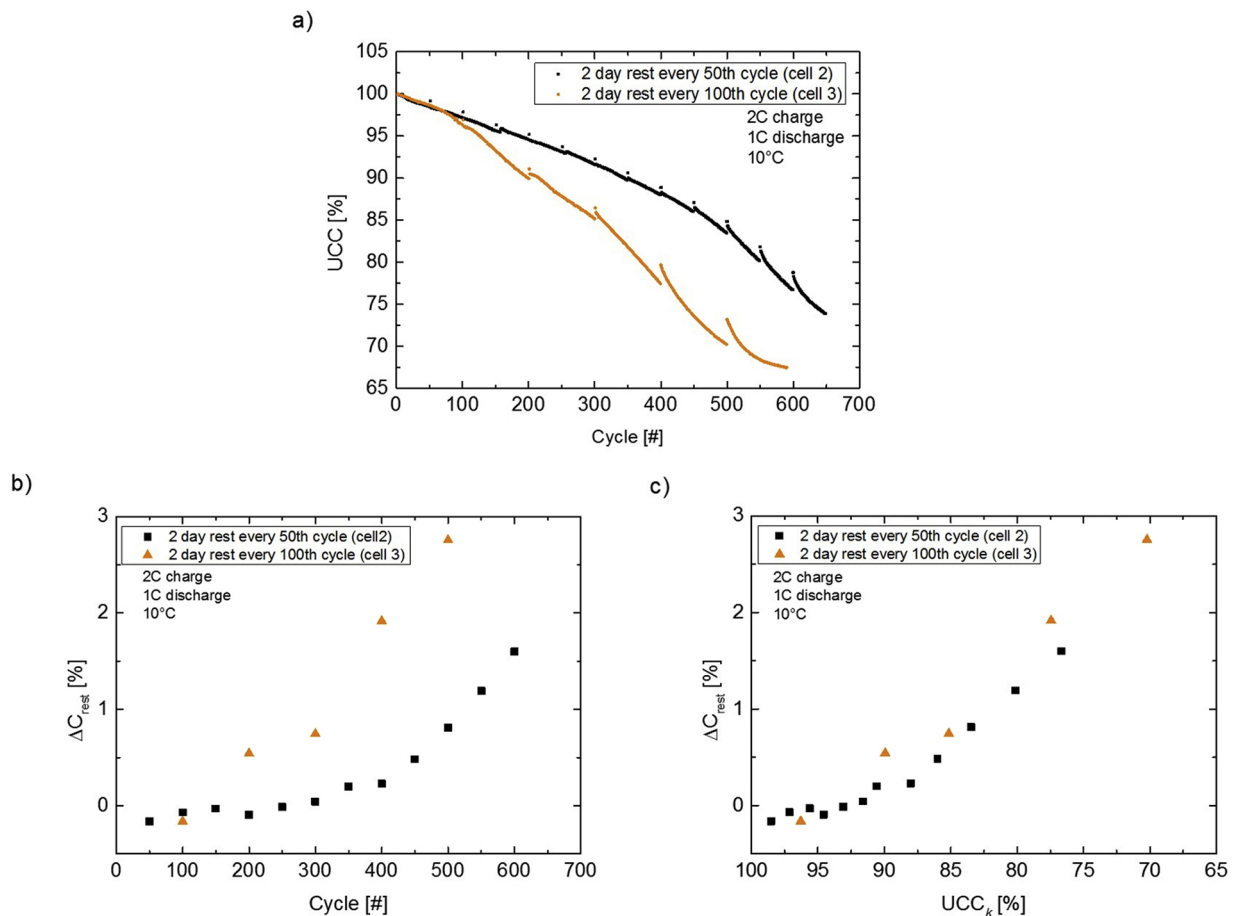


Fig. 7. Comparison of two cells which were cycled either 50 (cell 2) or 100 (cell 3) times between 2 day rest periods with a charge rate of 2 C and a discharge rate of 1 C at 10 °C. In a) the usable capacity over cycle number is shown. Moreover, the capacity increase during 2 day rest periods is shown over cycle number (b) and usable capacity before the rest period (c).

3.8. Specific capacity increase during rest periods

Fig. 8 shows the amount of recovered capacity during 2 day rest periods versus the amount of capacity lost during the 50 cycles since the rest period before. Three cell tests are compared which had only the 2 day rest periods at 100% SoC in common. A somewhat linear

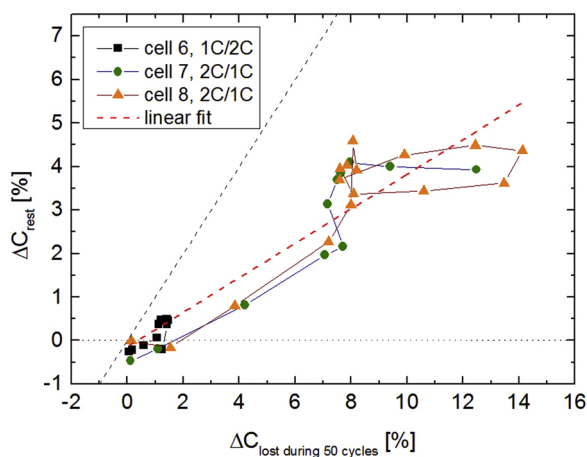


Fig. 8. Capacity increase during 2 day rest periods versus capacity loss over the 50 full cycles before the rest period. Test parameters for the cells are listed in Table 1. The dashed black line indicates full reversibility. The dashed red line is a fit through all data. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article).

correlation is observed between loss of capacity leading to the rest period and recovered capacity during the rest period for cells that were paused at 100% SoC. The red dashed line in Fig. 8 was fitted to the data with the following function:

$$y = (0.40 \pm 0.03)x - (0.14 \pm 0.19) \quad (8)$$

Thus, Fig. 8 shows that during a 2 day rest period after 50 cycles about 40% of “immobilized” metallic lithium, which was likely lost to lithium plating during the cycles before the rest period, can be recovered. Since the recovery process is not finished after 2 days, it is expected that longer rest periods would lead to an even more pronounced lithium recovery. Furthermore, calendar aging and lithium loss into the anode overhang are superimposed effects during the rest periods. Thus, the actual recoverable amount of metallic lithium might be even larger.

4. Conclusion

In this work, the effect of prolonged rest periods on the available capacity of automotive lithium ion cells during high current cyclic aging tests was investigated. It is demonstrated that the aging behavior can be influenced by prolonged rest periods as well as regular state of health characterizations which involve longer rest periods. A cell that has 2 day rest periods every 50 cycles achieved twice the number of full cycles before reaching 80% usable cell capacity when compared to a cell that is continuously cycled without any rest periods. Furthermore, it is shown that long rest periods can directly increase the available capacity in the following cycles by up to 7% usable cell capacity. The

occurrence of this capacity increase depends on the utilization of the cell. Higher current rates and lower temperatures lead to faster capacity loss and increased capacity recovery during rest periods. Thus, a kind of moderate temperature lithium plating caused by high current rates is a likely cause. This theory was strengthened using voltage relaxation tests which indicated lithium plating at the tested current rates. Since the rest periods in the cycle tests were conducted at 100% SoC the anode overhang effect can be excluded as the cause for the observed capacity increase. This, together with the almost negligible temperature influence on the capacity increase during rest periods, lead to the proposed microscopic model in which the capacity increase during rest periods is assumed to be limited by the electrical conductivity of a passivating layer that encapsulates the metallic lithium plating. Only when an electron moves through this passivating layer, the corresponding lithium ion can follow. With this model the exponential time dependency of the capacity increase during rest periods can be explained as observed in this work and by others. Furthermore, it was found that 2 day rest periods every 50th cycle lead to slower cell aging when compared to 2 day rest periods every 100th cycle. Lithium plating is believed to be a self-enforcing process and thus, regular rest periods can decrease the negative effect of lithium plating in the long run. Further analytical investigations into this effect are necessary to prove the proposed microscopic model. Until then this behavior needs to be kept in mind when designing accelerated cyclic cell aging tests and deriving aging models from tests which have regular state of health characterizations. Purely empirical models will likely fail to predict realistic usage scenarios correctly when based on accelerated aging tests. Only models which take the underlying physical processes into account promise a correct prediction.

Disclaimer

The results, opinions and conclusions expressed in this thesis are not necessarily those of Volkswagen Aktiengesellschaft.

Acknowledgement

We would like to thank Mirko Poyda for his support with the cell testing.

Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:<https://doi.org/10.1016/j.est.2019.02.015>.

References

- [1] J. Vetter, P. Novák, M.R. Wagner, C. Veit, K.-C. Möller, J.O. Besenhard, M. Winter, M. Wohlfahrt-Mehrens, C. Vogler, A. Hammouche, Ageing mechanisms in lithium-ion batteries, *J. Power Sources* 147 (1–2) (2005) 269–281.
- [2] P. Ramadass, B. Haran, P.M. Gomadam, R. White, B.N. Popov, Development of first principles capacity fade model for li-ion cells, *J. Electrochem. Soc.* (2) (2004) 151 p. A196.
- [3] M. Reichert, D. Andre, A. Rösmann, P. Janssen, H.-G. Brems, D.U. Sauer, S. Passerini, M. Winter, Influence of relaxation time on the lifetime of commercial lithium-ion cells, *J. Power Sources* 239 (2013) 45–53.
- [4] A. Eddahech, O. Briat, J.-M. Vinassa, Lithium-ion battery performance improvement based on capacity recovery exploitation, *Electrochim. Acta* 114 (2013) 750–757.
- [5] B. Gyenes, D.A. Stevens, V.L. Chevrier, J.R. Dahn, Understanding anomalous behavior in coulombic efficiency measurements on li-ion batteries, *J. Electrochem. Soc.* 162 (3) (2014) A278–A283.
- [6] M. Lewerenz, J. Münnix, J. Schmalstieg, S. Käbitz, M. Knips, D.U. Sauer, Systematic aging of commercial LiFePO₄/Graphite cylindrical cells including a theory explaining rise of capacity during aging, *J. Power Sources* 345 (2017) 254–263.
- [7] J. Wilhelm, S. Seidlmayer, P. Keil, J. Schuster, A. Kriele, R. Gilles, A. Jossen, Cycling capacity recovery effect, *J. Power Sources* 365 (2017) 327–338.
- [8] M. Lewerenz, D.U. Sauer, Evaluation of cyclic aging tests of prismatic automotive LiNiMnCoO₂-Graphite cells considering influence of homogeneity and anode overhang, *J. Energy Storage* 18 (2018) 421–434.
- [9] M. Lewerenz, G. Fuchs, L. Becker, D.U. Sauer, Irreversible calendar aging and quantification of the reversible capacity loss caused by anode overhang, *J. Energy Storage* 18 (2018) 149–18159.
- [10] M. Lewerenz, P. Dechent, D.U. Sauer, Investigation of capacity recovery during rest period at different states-of-charge after cycle life test for prismatic Li(Ni₁/3Mn₁/3Co₁/3)O₂-graphite cells, *J. Energy Storage* 21 (2019) 680–690.
- [11] M. Petzl, M. Kasper, M.A. Danzer, Lithium plating in a commercial lithium-ion battery – a low-temperature aging study, *J. Power Sources* 275 (2015) 799–807.
- [12] T. Waldmann, M. Wohlfahrt-Mehrens, Effects of rest time after Li plating on safety behavior—ARC tests with commercial high-energy 18650 Li-ion cells, *Electrochim. Acta* 230 (2017) 454–460.
- [13] M. Wünsch, J. Kaufman, D.U. Sauer, Investigation of the influence of different bracing of automotive pouch cells on cyclic lifetime and impedance spectra, *J. Energy Storage* 21 (2019) 149–155.
- [14] S. Schindler, M. Bauer, M. Petzl, M.A. Danzer, Voltage relaxation and impedance spectroscopy as in-operando methods for the detection of lithium plating on graphitic anodes in commercial lithium-ion cells, *J. Power Sources* 304 (2016) 170–180.
- [15] M. Broussely, S. Herreyre, P. Biensan, P. Kaszlejna, K. Nechev, R. Staniewicz, Aging mechanism in Li ion cells and calendar life predictions, *J. Power Sources* 97–98 (2001) 13–21.
- [16] M. Petzl, M.A. Danzer, Nondestructive detection, characterization, and quantification of lithium plating in commercial lithium-ion batteries, *J. Power Sources* 254 (2014) 80–87.
- [17] J.C. Burns, D.A. Stevens, J.R. Dahn, In-situ detection of lithium plating using high precision coulometry, *J. Electrochem. Soc.* 162 (6) (2015) A959–A964.
- [18] M. Lewerenz, A. Warnecke, D.U. Sauer, Introduction of capacity difference analysis (CDA) for analyzing lateral lithium-ion flow to determine the state of covering layer evolution, *J. Power Sources* 354 (2017) 157–166.
- [19] V. Zinth, C. Lüders, M. Hofmann, J. Hattendorff, I. Buchberger, S. Erhard, J. Rebelo-Kornmeier, A. Jossen, R. Gilles, Lithium plating in lithium-ion batteries at sub-ambient temperatures investigated by in situ neutron diffraction, *J. Power Sources* 271 (2014) 152–159.
- [20] M. Klett, P. Svens, C. Tengstedt, A. Seyeux, J. Świątowska, G. Lindbergh, R. Wreland Lindström, Uneven film formation across depth of porous graphite electrodes in cycled commercial li-ion batteries, *J. Phys. Chem. C* 119 (1) (2014) 90–100.
- [21] E. Peled, S. Menkin, Review—SEI, *J. Electrochem. Soc.* 164 (7) (2017) A1703–A1719.
- [22] M. Ecker, P. Shafiei Sabet, D.U. Sauer, Influence of operational condition on lithium plating for commercial lithium-ion batteries – electrochemical experiments and post-mortem-analysis, *Appl. Energy* 206 (2017) 934–946.
- [23] S.F. Schuster, T. Bach, E. Fleder, J. Müller, M. Brand, G. Sextl, A. Jossen, Nonlinear aging characteristics of lithium-ion cells under different operational conditions, *J. Energy Storage* 1 (2015) 44–53.
- [24] X.-G. Yang, Y. Leng, G. Zhang, S. Ge, C.-Y. Wang, Modeling of lithium plating induced aging of lithium-ion batteries, *J. Power Sources* 360 (2017) 28–40.
- [25] M. Klett, R. Eriksson, J. Groot, P. Svens, K. Ciosek Högström, R.W. Lindström, H. Berg, T. Gustafson, G. Lindbergh, K. Edström, Non-uniform aging of cycled commercial LiFePO₄/graphite cylindrical cells revealed by post-mortem analysis, *J. Power Sources* 257 (2014) 126–137.
- [26] M. Lewerenz, A. Warnecke, D.U. Sauer, Post-mortem analysis on LiFePO₄/Graphite cells describing the evolution & composition of covering layer on anode and their impact on cell performance, *J. Power Sources* 369 (2017) 122–132.